

The Parameter-Precompactness Criterion for Strong Bohr Approximation

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Abstract

We give a complete parameter-side answer on the canonical domain $I = \mathbb{R}^n$ to the converse question following Proposition 2.24 of Chávez–Khalil–Kostić–Pinto. For a family $\mathcal{B}$ of subsets of a Banach space X , every continuous Bohr $\mathcal{B}$ -almost periodic function on $\mathbb{R}^n \times X$ is strongly $\mathcal{B}$ -almost periodic if and only if every member of $\mathcal{B}$ is relatively compact. Thus for the family of all bounded subsets the universal implication holds exactly when X is finite-dimensional. The same characterization holds on the full-dimensional convex polyhedral cones used in the source. In contrast, when $n \geq 2$, diagonal almost periods alone do not imply strong almost periodicity even for a bounded scalar function with no parameter variable.

Source Question

After Proposition 2.24(iv)–(v) of [1], the authors ask:

Suppose that $\emptyset \neq I \subseteq \mathbb{R}^n$ and $F : I \times X \rightarrow Y$ is Bohr $\mathcal{B}$ -almost periodic (or Bohr $(\mathcal{B}, I \cap \Delta_n)$ -almost periodic). What conditions ensure the strong $\mathcal{B}$ -almost periodicity of F ?

Concerning Proposition 2.24(iv)–(v), it is natural to ask the following: Suppose that $\emptyset \neq I \subseteq \mathbb{R}^n$ and $F : I \times X \rightarrow Y$ is a Bohr $\mathcal{B}$ -almost periodic (Bohr $(\mathcal{B}, I \cap \Delta_n)$ -almost periodic) function. What conditions ensure the strong $\mathcal{B}$ -almost periodicity of $F(\cdot; \cdot)$? After proving Theorem 2.36 it will be clear from a combination with Proposition 2.24(iv) that the notion of strong Bohr $\mathcal{B}$ -almost periodicity for continuous functions $F : I \rightarrow Y$ coincides with the notion of Bohr $\mathcal{B}$ -almost periodicity, provided that I is a convex polyhedral; as a simple consequence of the last mentioned theorem, we also have that the uniform convergence of a sequence of scalar-valued trigonometric polynomials on a convex polyhedral in $\mathbb{R}^n$ always implies the uniform convergence of this sequence on the whole space $\mathbb{R}^n$ (in the present state of our knowledge, we really do not know whether this result was known before stating above).

For $I = \mathbb{R}^n$, Bohr $\mathcal{B}$ -almost periodicity means that for every $B \in \mathcal{B}$ and $\varepsilon > 0$ there is $L > 0$ such that every ball of radius L contains a translation τ satisfying

$$\sup_{t \in \mathbb{R}^n, x \in B} \|F(t + \tau, x) - F(t, x)\|_Y \leq \varepsilon.$$

Strong $\mathcal{B}$ -almost periodicity means that, for every $B \in \mathcal{B}$, F is the uniform limit on $\mathbb{R}^n \times B$ of polynomials

$$P(t, x) = \sum_{j=1}^r e^{i\langle \lambda_j, t \rangle} c_j(x), \quad \lambda_j \in \mathbb{R}^n, \quad c_j \in C(X; Y).$$

Proof Intuition

There are two independent obstructions. On the parameter side, compactness turns F into the ordinary Banach-valued Bohr function $t \mapsto F(t, \cdot)|_K \in C(K; Y)$. Its vector-valued trigonometric approximants have coefficients in $C(K; Y)$, and Dugundji's theorem extends those coefficients from K to all of X .

This is sharp. Any non-precompact parameter set contains a uniformly separated sequence. Disjoint bumps around that sequence can carry the successive modes e^{imt_1} . The resulting function has the exact common period 2π but no finite-frequency polynomial can approximate all the modes: a character mean gives error at least one.

Diagonal translations introduce a different, purely temporal obstruction. They do not see the transverse coordinate $t_1 - t_2$. The bounded function $e^{i(t_1 - t_2)^2}$ is therefore exactly invariant under every diagonal shift, but it cannot be a uniform limit of trigonometric polynomials because its restriction e^{is^2} is not uniformly continuous.

Exact Euclidean-Domain Characterization

Theorem 1 (parameter-precompactness criterion). *Let $n \geq 1$, let X be a complex Banach space, and let $\mathcal{B}$ be a nonempty family of nonempty subsets of X . The following are equivalent.*

1. *Every $B \in \mathcal{B}$ is relatively compact in X .*
2. *For every complex Banach space Y , every continuous Bohr $\mathcal{B}$ -almost periodic function $F : \mathbb{R}^n \times X \rightarrow Y$ is strongly $\mathcal{B}$ -almost periodic.*
3. *Statement (2) holds for scalar-valued functions $Y = \mathbb{C}$.*

When these conditions hold and the members of $\mathcal{B}$ are bounded, Bohr and strong $\mathcal{B}$ -almost periodicity are equivalent.

Proof. Assume (1), fix $B \in \mathcal{B}$, and put $K = \overline{B}$. Then K is compact. For

$$G_K(t) = F(t, \cdot)|_K \in C(K; Y), \quad t \in \mathbb{R}^n,$$

the map $G_K : \mathbb{R}^n \rightarrow C(K; Y)$ is norm-continuous. Indeed, joint continuity of F at the points of $\{t_0\} \times K$, followed by a finite-cover argument on K , gives

$$\sup_{x \in K} \|F(t, x) - F(t_0, x)\|_Y \longrightarrow 0 \quad (t \rightarrow t_0).$$

For each fixed t, τ , continuity in x and density of B in K give

$$\sup_{x \in B} \|F(t + \tau, x) - F(t, x)\|_Y = \sup_{x \in K} \|F(t + \tau, x) - F(t, x)\|_Y.$$

Hence the Bohr $\mathcal{B}$ estimate for B says exactly that G_K is a $C(K; Y)$ -valued Bohr almost periodic function.

The Banach-valued Bohr approximation theorem recalled in Section 1.1 of [1] gives trigonometric polynomials

$$Q_q(t) = \sum_{j=1}^{r_q} e^{i\langle \lambda_{q,j}, t \rangle} a_{q,j}, \quad a_{q,j} \in C(K; Y),$$

such that

$$\sup_{t \in \mathbb{R}^n} \|G_K(t) - Q_q(t)\|_{C(K;Y)} \longrightarrow 0.$$

The set K is closed in the metric space X , and Y is locally convex. Dugundji's extension theorem [2] extends every $a_{q,j}$ to a continuous map $c_{q,j} : X \rightarrow Y$. Therefore

$$P_q^B(t, x) = \sum_{j=1}^{r_q} e^{i\langle \lambda_{q,j}, t \rangle} c_{q,j}(x)$$

has the exact form required in the source and converges uniformly to F on $\mathbb{R}^n \times K$, hence on $\mathbb{R}^n \times B$. Since B was arbitrary, F is strongly $\mathcal{B}$ -almost periodic. This proves (1) $\Rightarrow$ (2), while (2) $\Rightarrow$ (3) is immediate.

We prove the contrapositive of (3) $\Rightarrow$ (1). Suppose that some $B_0 \in \mathcal{B}$ is not relatively compact. Since X is complete, B_0 is not totally bounded. Thus there are $\delta > 0$ and points $x_m \in B_0$ with

$$\|x_m - x_k\| \geq \delta \quad (m \neq k).$$

Define

$$\phi_m(x) = \max\{0, 1 - 3\|x - x_m\|/\delta\}.$$

The supports lie in the radius- $\delta/3$ balls around the x_m and are pairwise separated. The family (ϕ_m) is locally finite, and at most one term is nonzero at any point. Set

$$F(t, x) = \sum_{m=1}^{\infty} \phi_m(x) e^{imt_1}. \quad (1)$$

Then F is jointly continuous and $|F| \leq 1$. Every $\tau \in \mathbb{R}^n$ with $\tau_1 \in 2\pi\mathbb{Z}$ is an exact period. These periods are relatively dense: one changes the first coordinate of any point to its nearest multiple of 2π , moving by at most π . Thus F is Bohr $\mathcal{B}$ -almost periodic for every family $\mathcal{B}$.

Let

$$P(t, x) = \sum_{j=1}^r e^{i\langle \lambda_j, t \rangle} c_j(x)$$

be any allowed trigonometric polynomial. Choose m such that $me_1 \notin \{\lambda_1, \dots, \lambda_r\}$. Since $F(t, x_m) = e^{imt_1}$, multiplication by e^{-imt_1} and averaging over $Q_T = [-T, T]^n$ gives

$$\begin{aligned} & \frac{1}{(2T)^n} \int_{Q_T} e^{-imt_1} (F(t, x_m) - P(t, x_m)) dt \\ &= 1 - \sum_{j=1}^r c_j(x_m) \frac{1}{(2T)^n} \int_{Q_T} e^{i\langle \lambda_j - me_1, t \rangle} dt \longrightarrow 1. \end{aligned}$$

Every nontrivial character has large-cube mean zero. The absolute value of the left side is at most the uniform error on $\mathbb{R}^n \times B_0$; hence

$$\sup_{t \in \mathbb{R}^n, x \in B_0} |F(t, x) - P(t, x)| \geq 1.$$

No strong approximation exists, contradicting (3). This proves the equivalence.

Finally, if the members of $\mathcal{B}$ are bounded, strong $\mathcal{B}$ -almost periodicity implies Bohr $\mathcal{B}$ -almost periodicity by Proposition 2.24(iv) of [1] (the domain is $\mathbb{R}^n$). The last assertion follows. $\square$

Corollary 1. *For the family of all compact subsets of X , Bohr and strong $\mathcal{B}$ -almost periodicity coincide on $\mathbb{R}^n$. For the family of all bounded subsets, the universal implication from Bohr to strong $\mathcal{B}$ -almost periodicity holds if and only if X is finite-dimensional.*

Proof. Compact sets are relatively compact. All bounded subsets of a Banach space are relatively compact exactly in finite dimension. $\square$

Convex Polyhedral Domains

The same criterion applies to the source's principal non-group domains.

Theorem 2. *Let $(v_1, \dots, v_n)$ be a basis of $\mathbb{R}^n$ and*

$$I = \left\{ \sum_{j=1}^n a_j v_j : a_j \geq 0 \right\}.$$

For a family $\mathcal{B}$ of subsets of a Banach space X , every continuous Bohr $\mathcal{B}$ -almost periodic map $F : I \times X \rightarrow Y$ is strongly $\mathcal{B}$ -almost periodic for every Banach target Y if and only if every $B \in \mathcal{B}$ is relatively compact.

Proof. For sufficiency, fix B and $K = \overline{B}$. As above, $G_K(t) = F(t, \cdot)|_K$ is a continuous Bohr almost periodic map $I \rightarrow C(K; Y)$. The convex-polyhedral consequence of Theorem 2.36 in [1], applied to the Banach target $C(K; Y)$, uniformly approximates G_K on I by $C(K; Y)$ -valued trigonometric polynomials. Dugundji extension of their coefficients gives the required strong approximants.

For necessity, let (x_m) and (ϕ_m) be the separated sequence and bumps from the proof of Theorem 1. Let ℓ_1 be the dual coordinate with $\ell_1(v_1) = 1$ and $\ell_1(v_j) = 0$ for $j > 1$, and put

$$F(t, x) = \sum_{m \geq 1} \phi_m(x) e^{im\ell_1(t)}, \quad t \in I.$$

This function is bounded and continuous. Its exact periods in I include all τ with $\ell_1(\tau) \in 2\pi\mathbb{Z}$. They are relatively dense in I : if $t^0 = \sum a_j v_j$, replace a_1 by its nearest nonnegative multiple of 2π , changing t^0 by at most $\pi\|v_1\|$.

If a finite trigonometric polynomial approximated F on $I \times B_0$, restricting at x_m to the ray $t = sv_1$, $s \geq 0$, would approximate e^{ims} by finitely many characters $e^{i\langle \lambda_j, v_1 \rangle s}$. Choose m outside that finite frequency set and average over $[0, T]$ after multiplication by e^{-ims} . The same character-mean argument gives error at least one. Thus non-precompact B_0 again contradicts the universal implication. $\square$

Diagonal Periods Do Not Suffice

Theorem 3. *Let $n \geq 2$. Even with no effective parameter variable, there is a bounded continuous scalar function on $\mathbb{R}^n$ that is Bohr $(\mathcal{B}, \Delta_n)$ -almost periodic but is not strongly $\mathcal{B}$ -almost periodic. Consequently no condition on the parameter family alone can reverse Proposition 2.24(v).*

Proof. Take $X = \{0\}$, $\mathcal{B} = \{\{0\}\}$, and

$$F(t) = e^{i(t_1 - t_2)^2}.$$

Every diagonal vector $(\tau, \dots, \tau)$ is an exact period, so the diagonal Bohr condition holds with zero error.

If F were strongly almost periodic, then its restriction to the line $t = (s, 0, \dots, 0)$ would be a uniform limit on $\mathbb{R}$ of ordinary trigonometric polynomials. Such a limit is uniformly continuous. But e^{is^2} is not uniformly continuous: for

$$a_k = \sqrt{2\pi k}, \quad b_k = \sqrt{2\pi k + \pi},$$

we have $|a_k - b_k| \rightarrow 0$ while $e^{ia_k^2} = 1$ and $e^{ib_k^2} = -1$. This contradiction proves the claim. $\square$

Verification Notes and Scope

Theorem 1 upgrades the earlier compact-versus-bounded example to an exact criterion for every parameter family on $\mathbb{R}^n$. Its necessity uses only completeness of X and the elementary equivalence between failure of total boundedness and the existence of a separated sequence. It is not special to ℓ_2 .

Theorem 2 covers the full-dimensional convex polyhedral cones explicitly used after Theorem 2.36 in the source. Arbitrary additive domains $I \subsetneq \mathbb{R}^n$ may have additional temporal extension obstructions and are not classified here. Theorem 3 completely rules out a parameter-only converse for diagonal periods in dimensions at least two.

Novelty Check

The run indexes and local arXiv source corpus were searched for arXiv:2012.00543, the exact converse question, and combinations of “strong $\mathcal{B}$ -almost periodic”, “Bohr $\mathcal{B}$ -almost periodic”, “relatively compact”, and “precompact”. Bounded web searches on 2026-08-09 checked the original paper and later work on metrical and Stepanov variants. They found the source and related generalizations, but no later answer, parameter-precompactness iff criterion, or separated-frequency counterexample matching the theorems above. This is a bounded novelty check, not a certification of originality.

Human-Review Recommendation

Treat Theorems 1–3 as a likely-valid complete classification for $I = \mathbb{R}^n$, with a convex-polyhedral extension. The key review points are the $C(K; Y)$ norm-continuity step, Dugundji extension of vector-valued coefficients, local finiteness of the separated bump family, the character-mean lower bound, and the exact scope of the source’s Theorem 2.36 used in Theorem 2. No conditional lemma remains.

References

- [1] A. Chávez, K. Khalil, M. Kostić, and M. Pinto, *Multi-dimensional almost periodic type functions and applications*, arXiv:2012.00543 (2020).
- [2] J. Dugundji, *An extension of Tietze’s theorem*, Pacific Journal of Mathematics 1 (1951), no. 3, 353–367.